Accelerating Directed Evolution

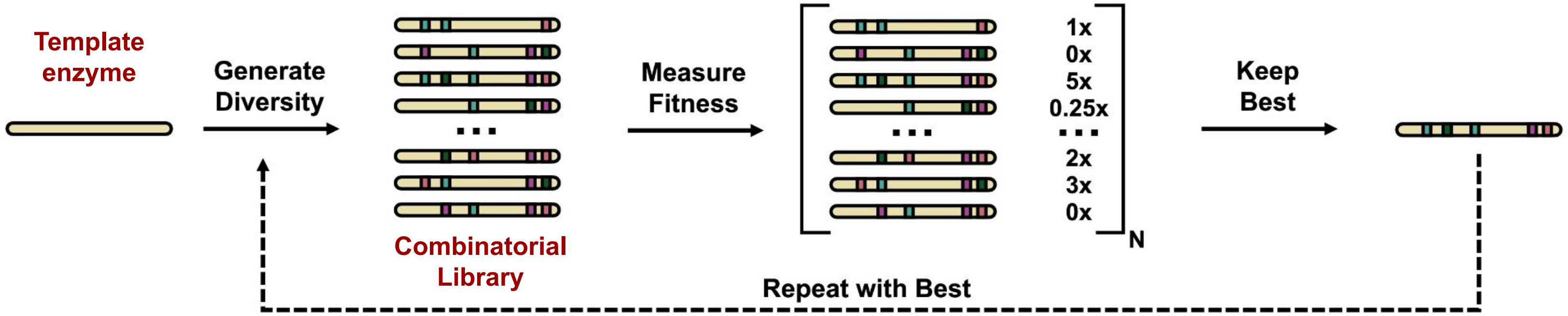
(with AI and more)

3rd April, 2024

Aditi Pathak

Directed Evolution

- Directed evolution functions by **harnessing natural evolution**, but on a **shorter timescale**
- Inspired by natural evolution, directed evolution leads to an **accumulation of beneficial mutations** via an iterative protocol of mutation and selection



Whitman et al. (2021)

The Reward

- **Finding optimal proteins** for increased enzyme activity, improved stability under harsh conditions, or altered substrate specificity

- **Bypassing knowledge gaps**

- **Wide applicability** to a variety of biomolecules like enzymes, proteins, and even RNA

The Challenge

- Finding the right selection pressure

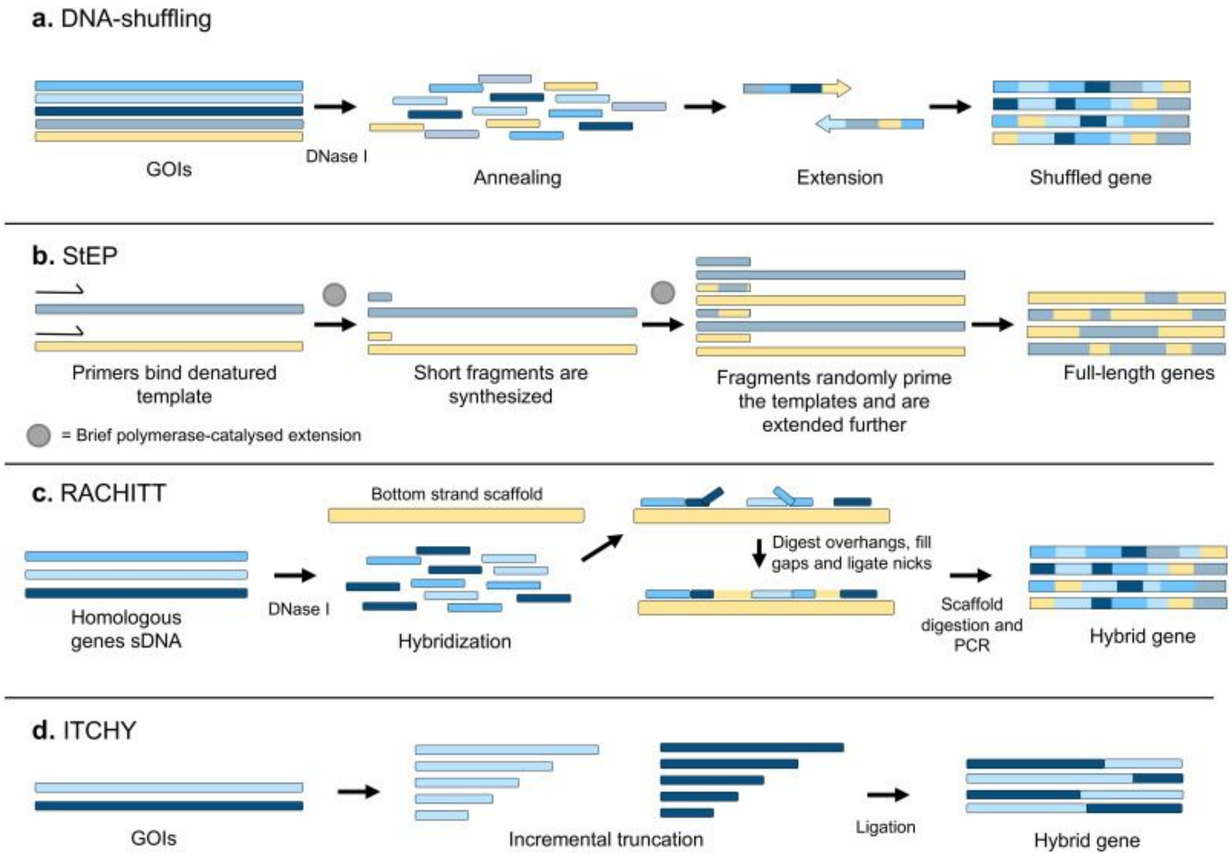
- Off-target effects

- Navigating the fitness landscape

- **Time and resource intensive**

For a protein with N number of residues, total possible combinations = 20^N

Recombination Techniques



Site-saturation and site-directed mutagenesis

Site-saturation mutagenesis allows the introduction of a large range of point-mutations at specific sites, while site-directed mutagenesis introduces a specific set of mutations

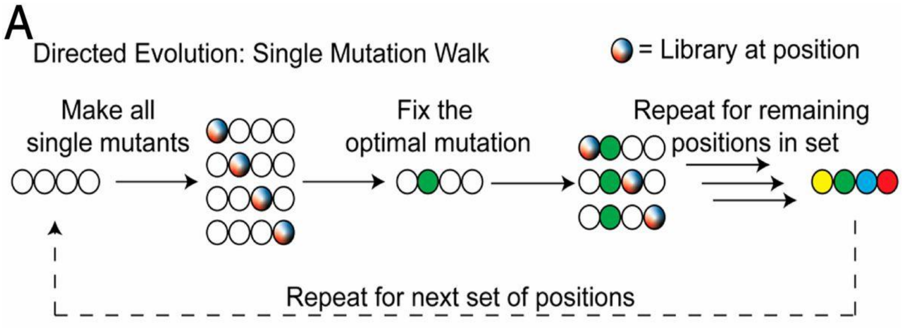
PNAS

RESEARCH ARTICLE | APPLIED BIOLOGICAL SCIENCES |

Machine learning-assisted directed protein evolution with combinatorial libraries

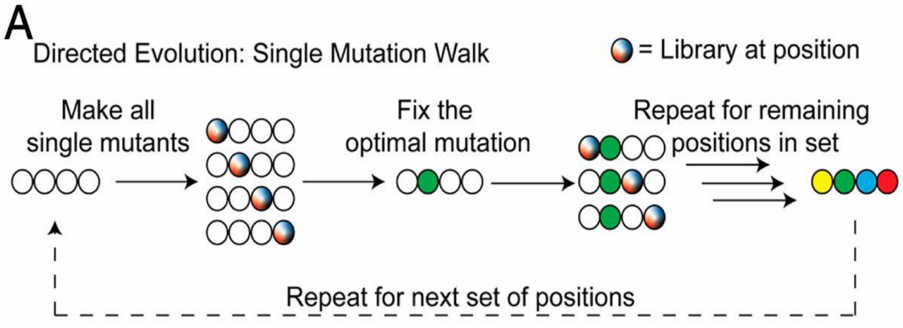
Zachary Wu, S. B. Jennifer Kan, Russell D. Lewis, Bruce J. Wittmann, and Frances H. Arnold

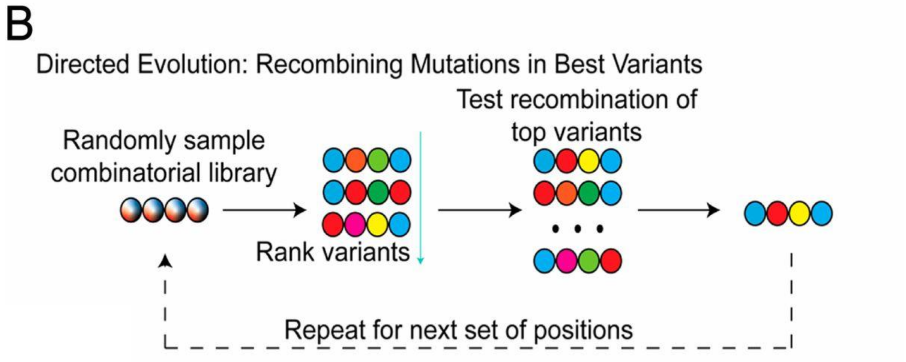
Conventional directed evolution



- In a conventional directed evolution experiment with **sequential single mutations**, the identification of optimal amino acids for **N positions in a set requires N rounds of evolution**

Conventional directed evolution



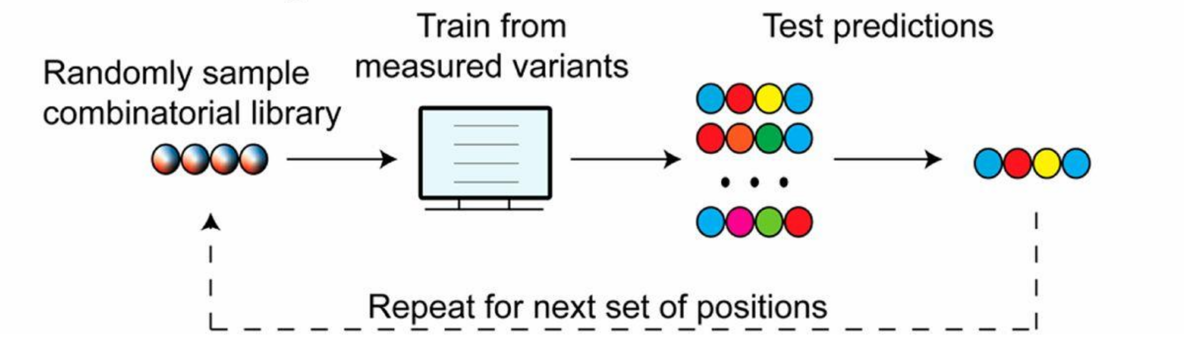


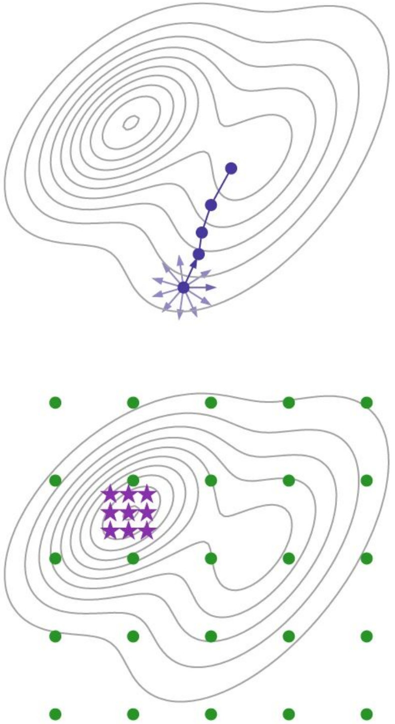
- In a conventional directed evolution experiment with **sequential single mutations**, the identification of optimal amino acids for **N positions in a set requires N rounds of evolution**

- An alternative approach is to **randomly sample** the combinatorial space and **recombine the best mutations found at each position** in a subsequent combinatorial library

Machine learning assisted directed evolution

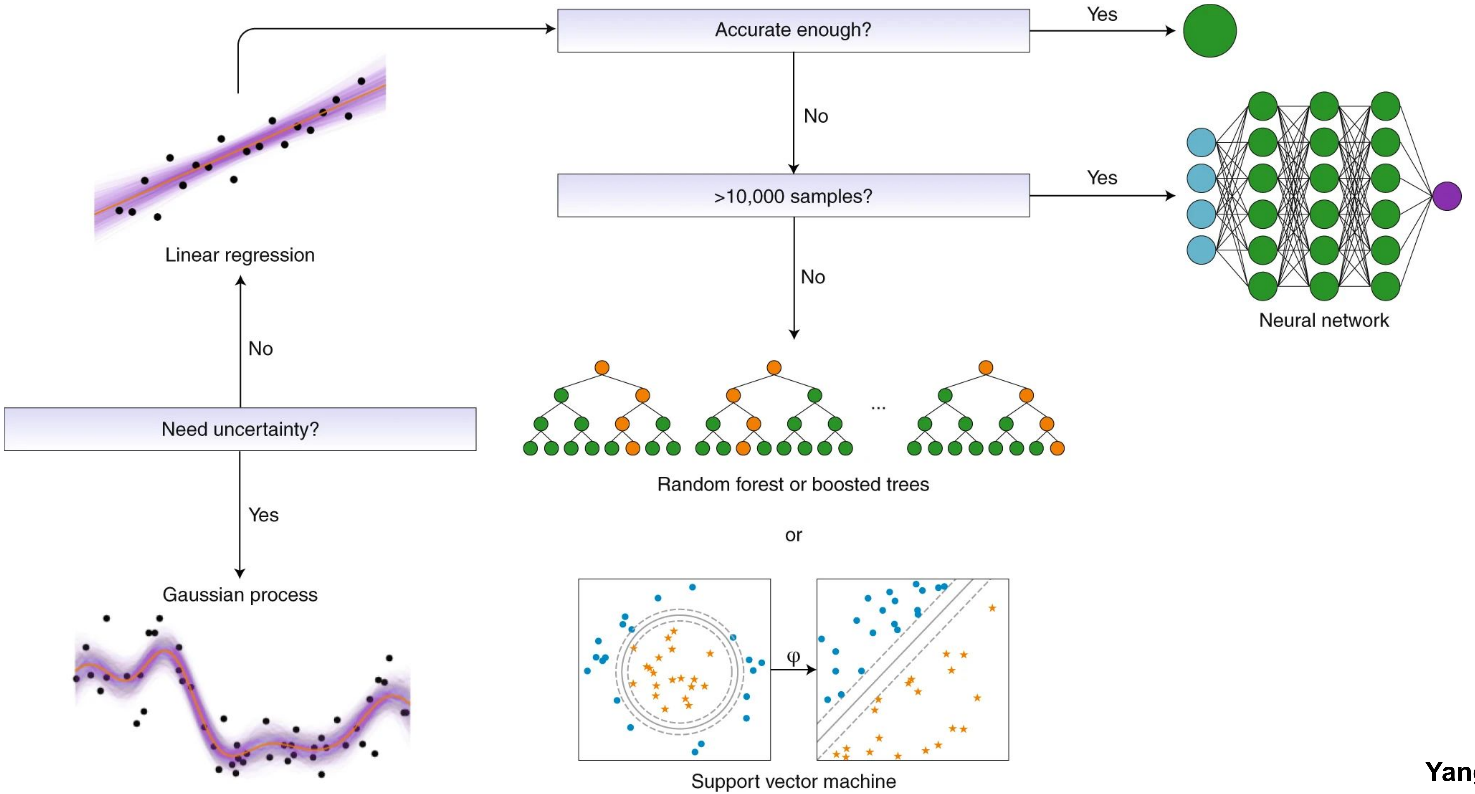
- Data from a **random sample of the combinatorial library**, the input library, are used to **train machine learning models**
- These models are used to **predict a smaller set of variants**, the predicted library, which can be encoded to test experimentally
- The **best-performing variant** is then used as the **parent sequence for the next round of evolution** with mutations at new positions.





Yang et al. (2019)

A general heuristic for choosing a machine-learning sequence–function model for proteins



Yang et al. (2019)

ML Models Tested

LinearRegression

LinearSVR

ARDRegression

BayesianRidge

ElasticNet

LassoLarsCV

KNeighborsRegressor

DecisionTreeRegressor

GradientBoostingRegressor

RandomForestRegressor

BaggingRegressor

AdaBoostRegressor

Multilayer Perceptron

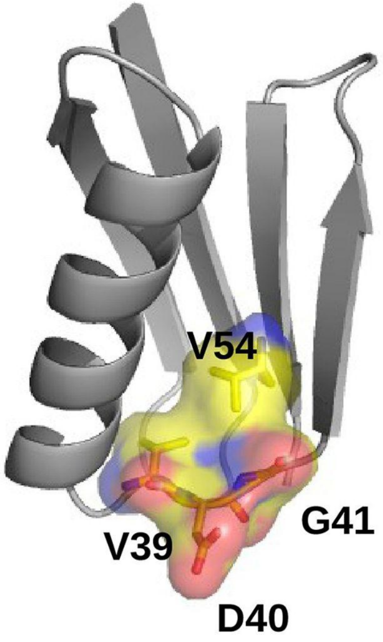
Validation on an empirical fitness landscape

- The ML-assisted approach was tested on the **large empirical fitness landscape** from Wu *et al.*

- Effects on **antibody binding** of mutations at **four positions** in **human GB1 binding protein** (theoretical library size, **20^4 = 160,000 variants**)

- The **fitness** of protein GB1 was defined as the **enrichment of folded protein bound to the antibody IgG-Fc** measured by coupling mRNA display with next-generation sequencing.

- Simulated single-mutation evolutionary walks starting from each of the **149,361 variants reported**



Wu et al. (2016)

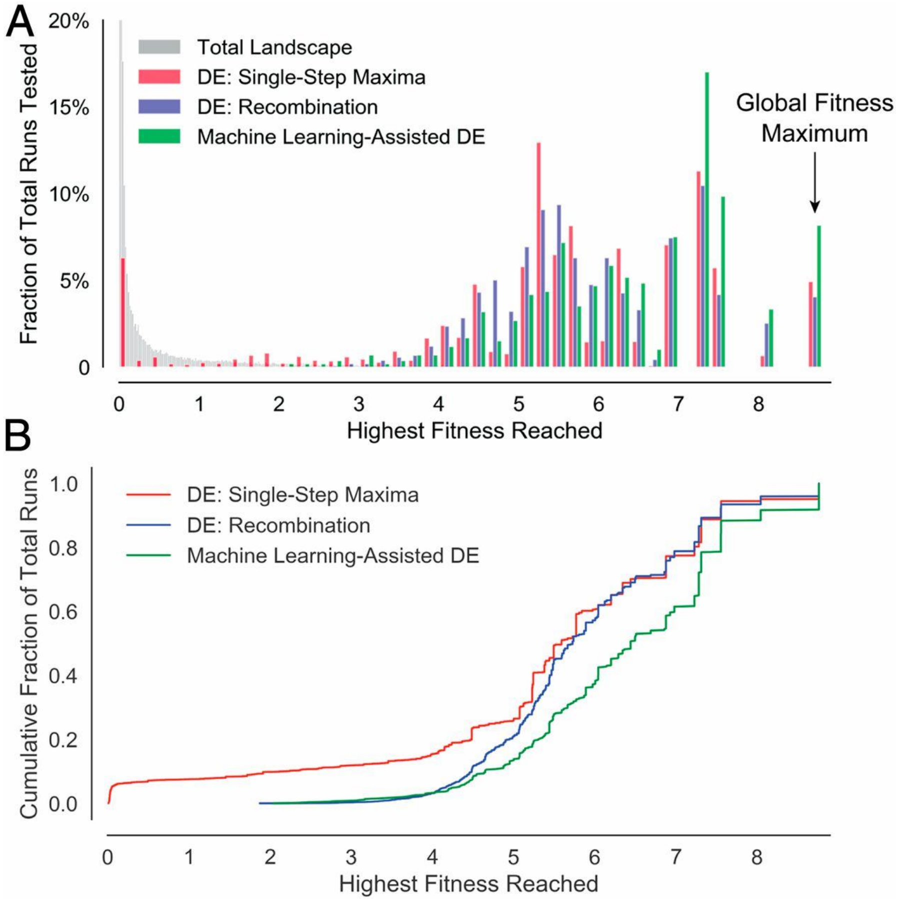
Validation on an empirical fitness landscape

- The distribution of fitness peaks found by **iterative site-saturation mutagenesis** from all labeled variants (149,361 of 204 possible covering four residues) is shown in red.

- The distribution of fitness peaks found by **10,000 recombination runs** with an average of 570 variants tested is shown in blue.

- The distribution of the highest fitnesses found from **600 runs of the machine learning assisted approach** is shown in green.

- A total of **570 variants are tested** in all approaches.



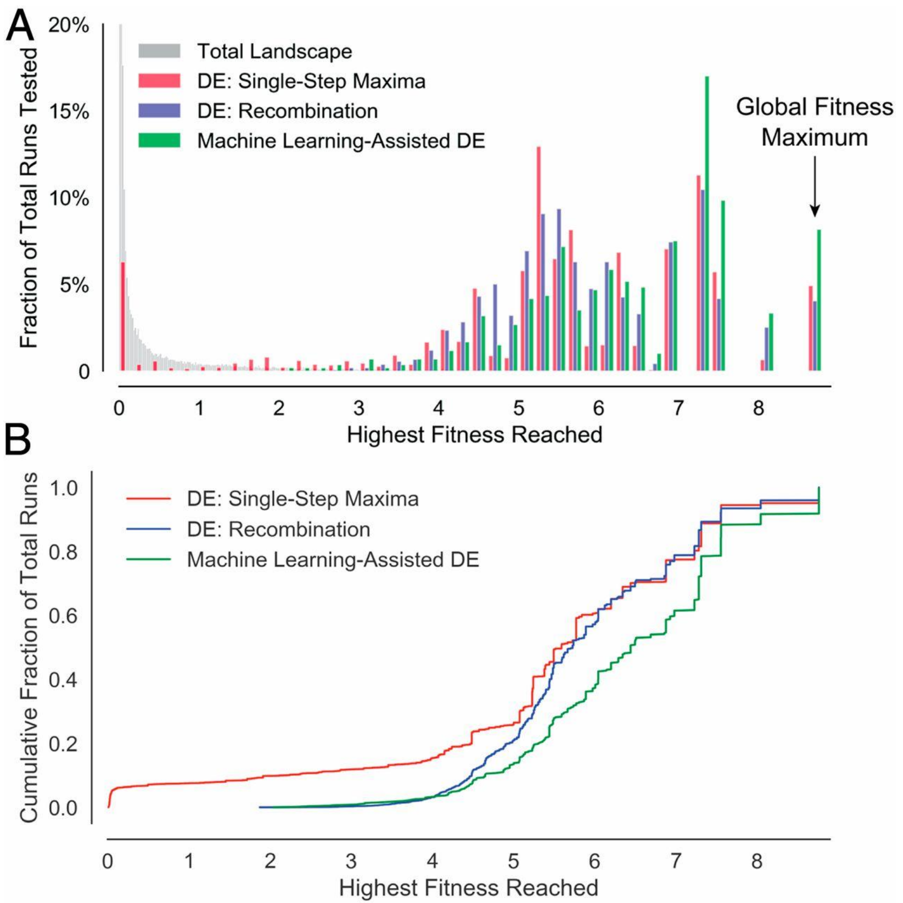
Validation on an empirical fitness landscape

With the same number of variants screened -

- ML-assisted evolution reaches the global **optimum fitness** value in **8.2% of 600 simulations**.
- **4.9%** of all starting sequences reach the same value through a **single mutant walk** and **4.0%** of **simulated recombination runs**.

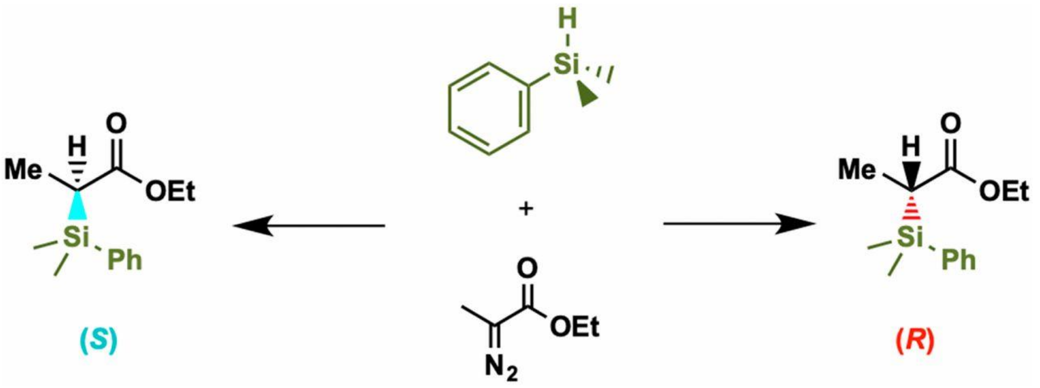
- The machine-learning approach requires **approximately 30% fewer variants** to achieve final results similar to the single-mutant walk with this analysis.

- **Accuracy** of the machine learning models as determined on a test set of 1,000 random variants not found in the training set **can be quite low** (Pearson's r = 0.41, SD = 0.17)
- However, this level of accuracy as measured by Pearson's r **appears to be sufficient to** guide evolution.



Evolution of enantiodivergent enzyme activity

- **Test case:** reaction of **phenyldimethyl silane with ethyl 2-diazopropanoate (Me-EDA**) catalyzed by a putative **nitric oxide dioxygenase (NOD)** from *Rhodothermus marinus*
- **Carbon–silicon bond formation** is a new-to-nature enzyme reaction
- Silicon has potential for **tuning the pharmaceutical properties** of bioactive molecules



- Because **enantiomers** of bioactive molecules can have **stark differences in their biological effects,** access to both is important

- Screening for enantioselectivity typically **requires long chiral separations to discover beneficial mutations** in a low-throughput screen

- Rma NOD with **mutations Y32K and V97L** catalyzes this reaction with **76% ee** for the **(S)-enantiomer** in whole-cell reactions

Enantiomeric Excess (ee)

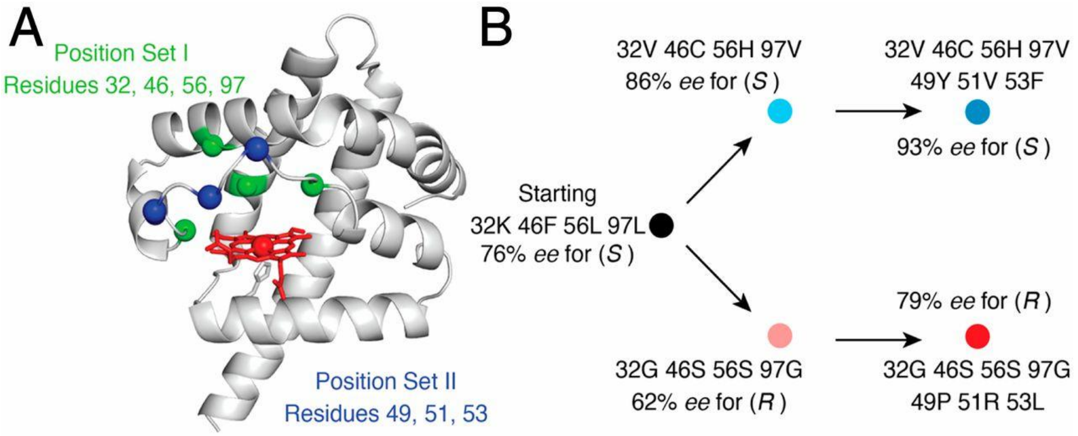
- Measurement of **purity for chiral substances**
- Reflects the **degree** to which a sample contains **one enantiomer in greater amounts than the other**
- A racemic mixture has an ee of 0%, while a single completely pure enantiomer has an ee of 100%

Evolution of enantiodivergent enzyme activity

Two sets of amino acid positions sampled :

- **Set I positions** were selected based on proximity to the **putative active site**
- **Set II positions** were selected based on their proximity to the putative **substrate entry channel**.

- 96-well plate was the smallest batch size
- Input data - 2 plates each round of evolution
- Tested predictions - 1 plate



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	Variants Input	Fraction of Total Library	Leave-One-Out CV Score	Predictions Tested
Position Set I	124	0.60 %	0.549	90 for (R) 90 for (S)
Position Set II from VCHV	155	8.3 %	0.630	90
Position Set II from GSSG	166	8.9 %	0.605	90

445 sequence–function relationships for model training
360 tested predicted variants

Solutions identified by ML approach can also be non-obvious

- If **only considering the last residue** the predicted library can be sampled from the top variants in the input library.
- However, for **each of the other three positions**, there is at least one mutation that is **not present in the top three input sequences**.

Table 1. Summary of the most (*S*)- and (*R*)-selective variants in the input and predicted libraries in set I (K32, F46, L56, L97)

Variant	Residue 32	46	56	97	Selectivity, % *ee* (enantiomer)
Input variants	Y	N	L	L	84 (*S*)
	C	S	V	L	83 (*S*)
	C	V	H	V	82 (*S*)
	C	R	S	G	56 (*R*)
	I	S	C	G	55 (*R*)
	N	V	R	I	47 (*R*)
Predicted variants	V	G	V	L	90 (*S*)
	C	F	N	L	90 (*S*)
	V*	C*	H*	V*	86 (*S*)
	G†	S†	S†	G†	62 (*R*)
	G	F	L	R	24 (*R*)
	H	C	S	R	17 (*R*)

*Parent sequence used for set II for (*S*)-selectivity.
†Parent sequence used for set II for (*R*)-selectivity.

Solutions identified by ML approach can also be non-obvious

- If **only considering the last residue** the predicted library can be sampled from the top variants in the input library.
- However, for **each of the other three positions**, there is at least one mutation that is **not present in the top three input sequences**.

- **Proline** is conserved at **residue 49** in two of the most (S)-selective variants.
- Proline is considered **unique for the conformational rigidity it confers**, and at first may seem structurally important, if not critical, for protein function.
- However, **tyrosine and arginine are also tolerated at position 49** with less than 1% loss in enantioselectivity

Table 2. Summary of the most (*S*)- and (*R*)-selective variants in the input and predicted libraries in position set II (P49, R51, I53)

Variant	Residue 49	Residue 51	Residue 53	Selectivity, % *ee* (enantiomer)	Cellular activity increase over KFLL
Input variants					
From VCHV	P*	R*	I*	86 (*S*)	—
	Y	V	F	86 (*S*)	—
	N	D	V	75 (*S*)	—
From GSSG	P†	R†	I†	62 (*R*)	—
	N	S	Y	56 (*R*)	—
	N	I	I	55 (*R*)	—
Predicted variants					
From VCHV	Y	V	V	93 (*S*)	2.8-fold
	P	V	I	93 (*S*)	3.2-fold
	P	V	V	87 (*S*)	3.1-fold
From GSSG	P	R	L	79 (*R*)	2.2-fold
	P	G	L	75 (*R*)	2.1-fold
	P	F	F	70 (*R*)	2.2-fold

Mutations that improve selectivity for the (*S*)-enantiomer appear in the background of [32V, 46C, 56H, 97V (VCHV)] and for the (*R*)-enantiomer are in [32G, 46S, 56S, 97G (GSSG)]. Activity increase over the starting variant, 32K, 46F, 56L, 97L (KFLL), is shown for the final variants.

*Parent sequence used for set II for (*S*)-selectivity.

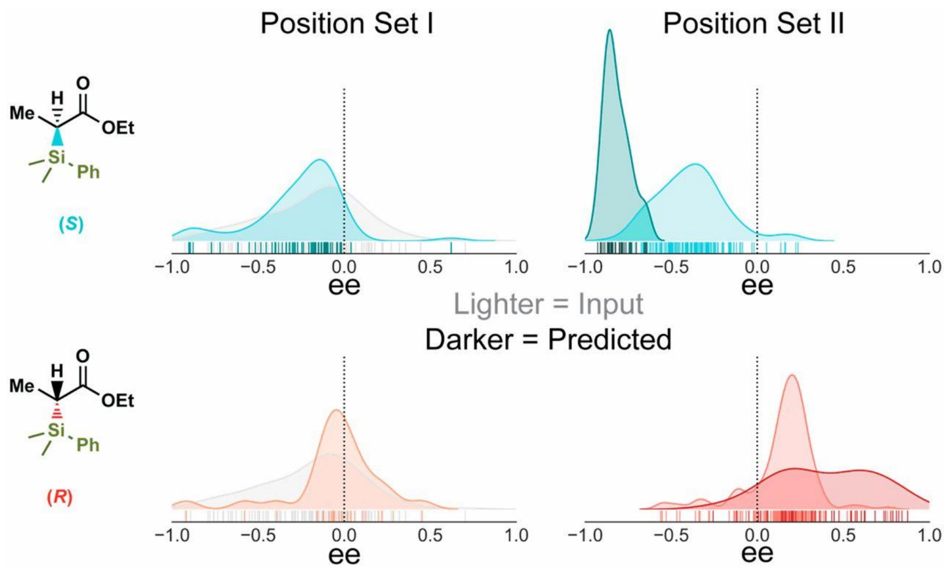
†Parent sequence used for set II for (*R*)-selectivity.

Machine learning predicts regions of sequence space enriched in function

An alternative representation of a library is a 1D distribution of fitness values sampled at random for each encoded library

- The distribution of random mutations made in the input libraries is **shifted toward the parent**
- In other words, random mutations made from an (R)-selective variant are more likely to be (R)-selective.

- The ML-approach is capable of **focusing its predictions** on areas of sequence space that are **enriched in high fitness**, as can be seen in the shift in distribution from input to predicted libraries



Discussion

- Machine learning can **quickly screen a full recombination library** in silico by using sequence–fitness relationships randomly sampled from the library

- Rather than relying on identifying beneficial single mutations, **epistatic interactions at the mutated positions can be modeled** by sampling the combinatorial sequence space directly

- ML-assisted approach can yield a **diverse set of solutions** -

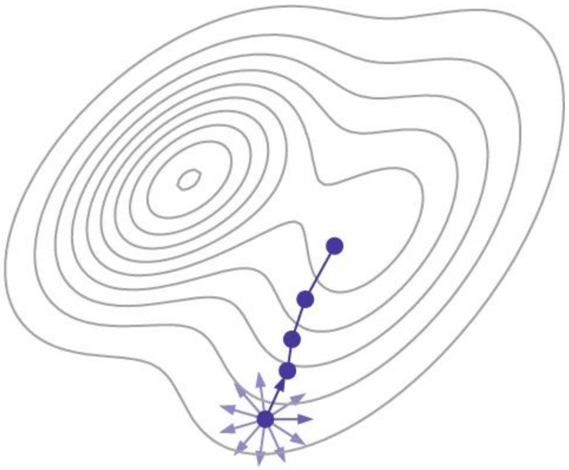
❖ Some of those variants may **satisfy other design requirements**

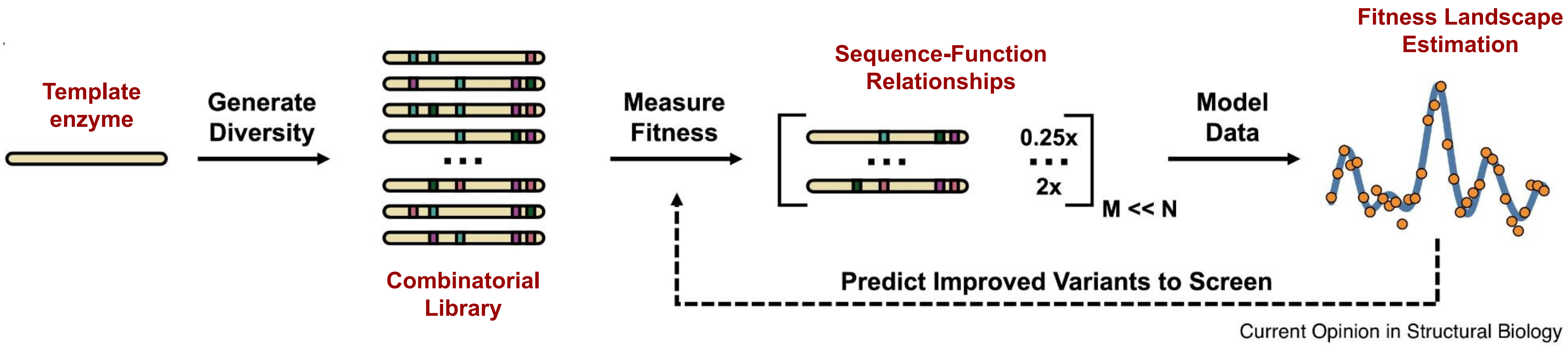
❖ Altered **substrate tolerance**

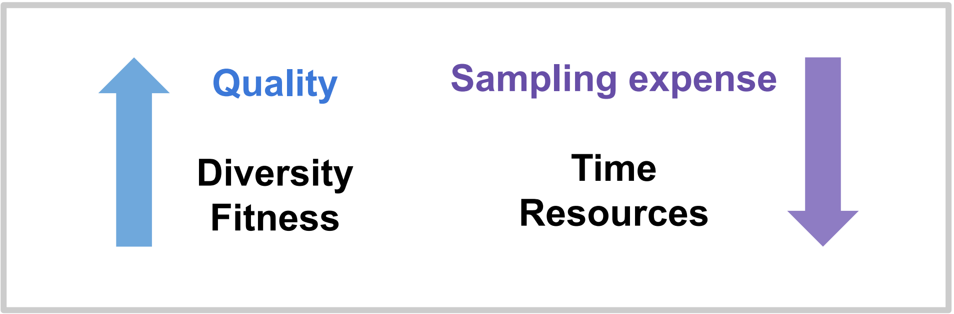
❖ Specific amino acid handles for **further protein modification**

❖ Sequence diversity for **intellectual property considerations**

Other Approaches to Assist Directed Evolution



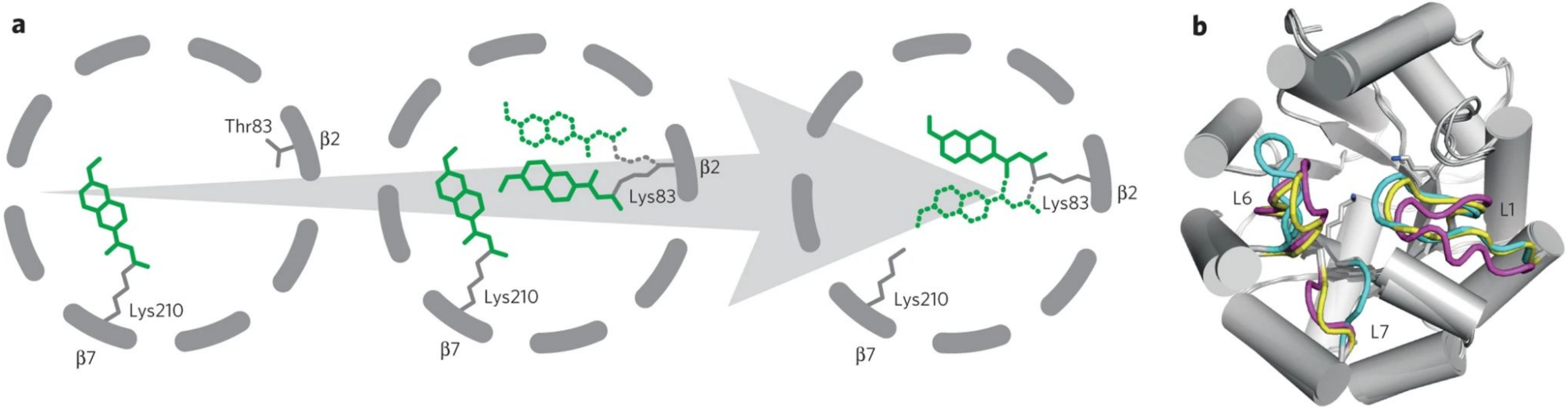

Template enzyme
Generate Diversity
Combinatorial Library
Measure Fitness
Sequence-Function Relationships
0.25x
2x
M << N
Model Data
Fitness Landscape Estimation
Predict Improved Variants to Screen
Current Opinion in Structural Biology


Quality
Diversity
Fitness
Sampling expense
Time
Resources

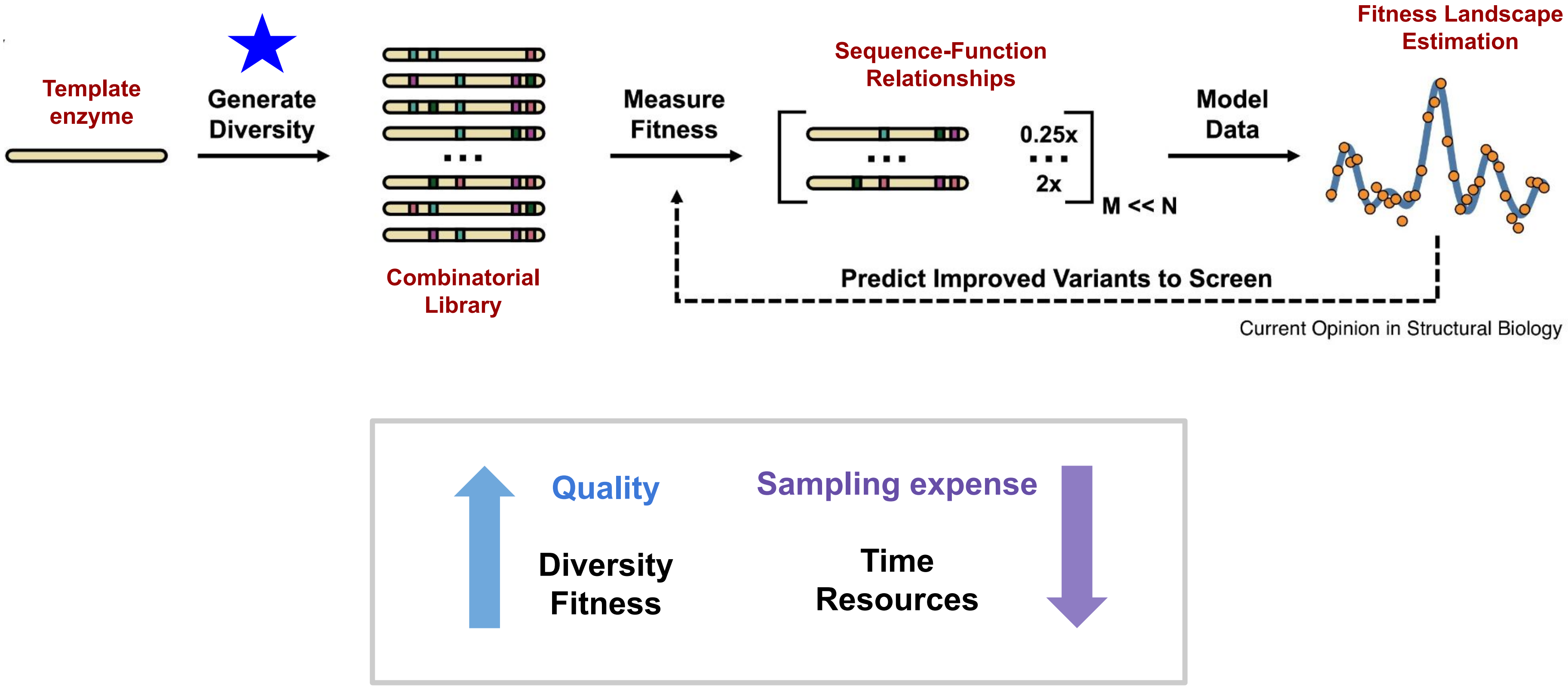
Article | Published: 09 June 2013

Evolution of a designed retro-aldolase leads to complete active site remodeling

Lars Giger, Sami Caner, Richard Obexer, Peter Kast, David Baker, Nenad Ban & Donald Hilvert



Evolutionary transformation of the computationally designed retro-aldolase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Current Opinion in Structural Biology
Quality
Diversity
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Sampling expense
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Resources

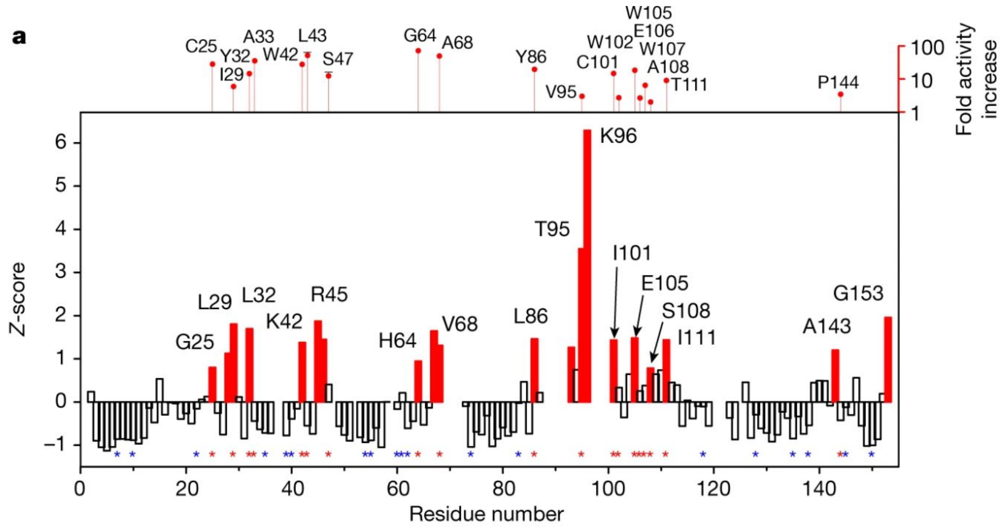
Article | Published: 05 October 2022

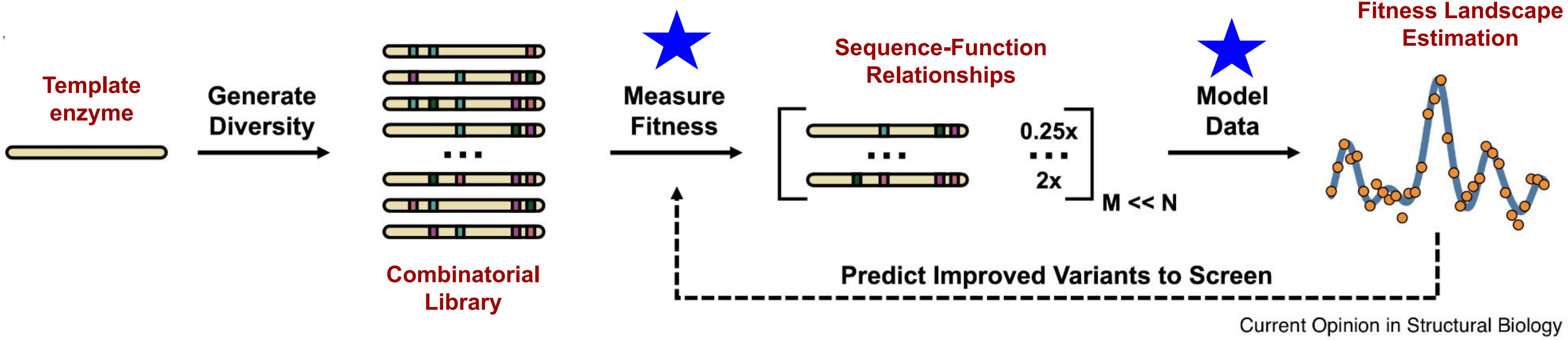
NMR-guided directed evolution

Sagar Bhattacharya, Eleonora G. Margheritis, Katsuya Takahashi, Alona Kulesha, Areetha D'Souza, Inhye Kim, Jennifer H. Yoon, Jeremy R. H. Tame, Alexander N. Volkov, Olga V. Makhlynets & Ivan V. Korendovych

- NMR-guided approach was used to **evolve a novel Kemp eliminase from Myoglobin**, a non-enzymatic haem protein.

- **Figure :**
  Backbone amide CSP of Mb(H64V) upon addition of 2 molar equivalents of 6-NBT.
  The red bars indicate the protein regions experiencing large CSP (Z>=1).

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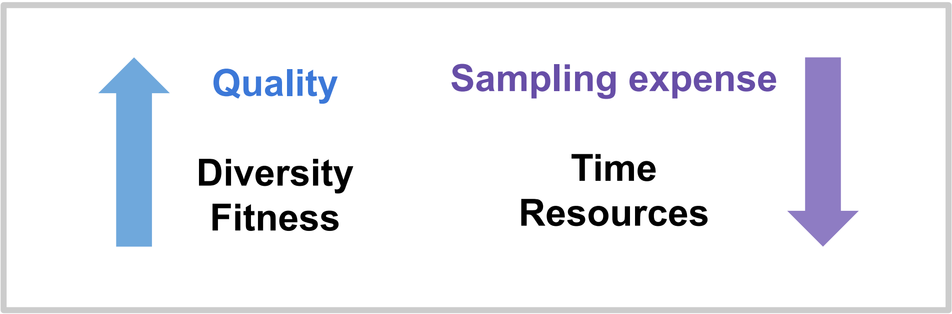
Quality
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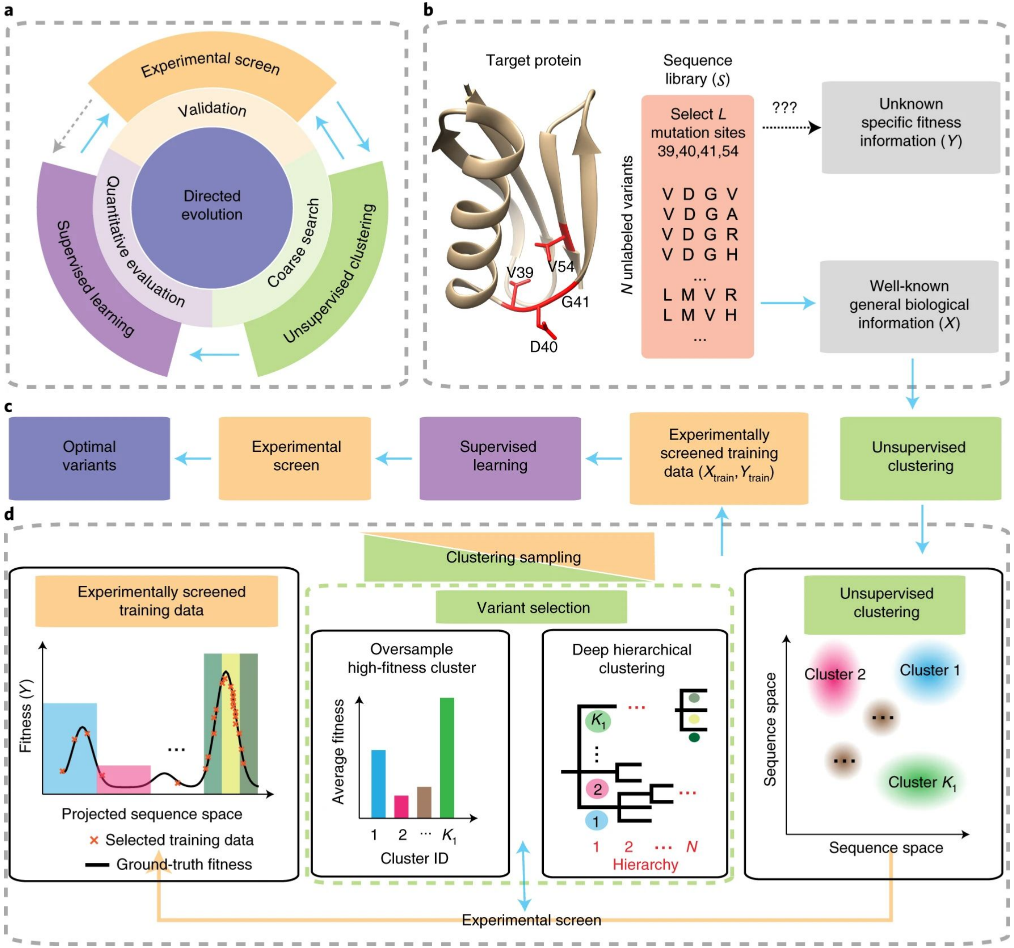
Article | Published: 09 December 2021

Cluster learning-assisted directed evolution

Yuchi Qiu, Jian Hu & Guo-Wei Wei



Current Opinion in Structural Biology

Advances in machine learning for directed evolution

Bruce J Wittmann[1], Kadina E Johnston[1], Zachary Wu[2 3], Frances H Arnold[1 2]